SPECIFICATION FOR SLIDING HANGAR DOOR_ NIHVA make	
DOOR CONSTRUCTION	The frame work for the elements shall be made out of hot rolled steel ISMC and other structural steel with welded/bolted connection satisfying all design requirement in strength satisfying IS Code provisions of IS 875-1987 (part-3) and deflection as well. All Leaves shall be inter- connected with suitable structural member along with rubber strike. Door leaves shall be of welded or bolted construction. Joints shall develop 100 percent of strength of framing members. Vertical members shall be continuous throughout the height of the door. When required, prepare splices to facilitate field assembly in accordance with standard practices. Frame and framing members shall be true to dimensions and square in all direction, no leaf shall be bowed, warped or out of line in the vertical or horizontal place of the door opening. Exposed welds and welds which interfere with the installation of various parts such as cover sheets shall be ground smooth.
PROTECTIVE COATING/ PAINTING ON STEEL COMPONENTS	1. One coat of Zinc Chromate Primer – 45 Microns 2. Two Coats of synthetic enamel paint – 60 Microns(@ 30 each)
CLADDING SYSTEM ON DOOR LEAVES	The Hangar Doors shall be cladded with approved vertically laid 0.5mm thick pre-coated corrugated galvalume sheet of make TATA Blue Scope/JSW/Essar. The Cladding System shall be fixed on the front face of the door, covering all structural elements of door leaf. * Insulation (Insulated panel) is optional
DOOR SEALS	All Vertical joints between leaves on adjacent tracks and the two end jambs are to be fitted with single EPDM wiper seals set in aluminum extrusions. At the jambs, suitable members are to be provided against which these seals mate. All seals shall be designed to resist draught and dust. Weather seals should be typically secured using zinc plated mild steel self – tapping or self-drilling screws.
BOTTOM WHEELS	Each door leaf is to be supported at the bottom on minimum of 2 Nos. Heavy duty double flanged steel track wheels of required diameter as per design requirement. The wheels assemblies shall be vertically adjustable, wheel assemblies shall be so that it permits removal of the wheel without removing the door leaf from its position on the rails. The machine wheel treads concentric with bearing. The distance between flanges shall not exceed the width of the rail by more than 3mm at the tread not more than 6mm at the edge of the flange. Wheel bearing Tapered rollers or spherical bearings, either internal or cartridge type, arranged so that both horizontal and vertical loads shall be transferred to the rail only through the bearing. Bearing shall be tightly sealed and equipped with high-pressure grease fittings.

TOP ROLLERS SYSTEMS	Each leaf is to be guided at top by min 2 Nos. horizontally running top roller assembly mounted on „ maintenance – free‟ sealed ball bearing for satisfactory performance under the designated load condition including live and lead loads, and rise and fall of the door truss upto L/360 (where "L" is the min. roof span dimensions). It shall be permanently factory-lubricated. Include Both Horizontal and Vertical rollers build into a frame which is connected in such a manner as to transmit the specified wind loads from the door to the hangar structure and to prevent disengagement of the door from the top of the guide.
TOP & BOTTOM TRACK	Top and Bottom Track of the Door shall be designed as per site requirement to withstand all type of dead and live load and smooth transmission of movement through motorized mechanism. Buffers: Heavy duty energy absorbing buffer stops are to be designed and to be provided at the extremities of the bottom track to cushion and stop the doors coming off the end of their tracks in an emergency situation.
MOTORIZED DRIVE DETAILS:	
A. (i) Electrical Motor capacity	The motor power in HP to be calculated according to door dimension, wind pressure and other required factors.
(ii) ElectricalOperation	The electrical operation of each door shall be carried out from a control panel mounted at the extreme end of the door openings inside the hangar building.
(a)	The person controlling the door shall be in full view of the door andin the vicinity of the door during the leaf movement.
(b)	Master panel is fitted with push button station. Thependant is mounted to the leading edge of the leaf on a cordabout 2 meters long that allows the operator to move from side toside to see where the door is being driven. The pendent is hung up on a hood when not in use.
(c)	The buttons themselves include "Open Mode" (as required) and "Emergency STOP" buttons. The "Open", "Close" push buttons are the dead man‟s type; the Emergency " STOP " button is a red mushroom-headed, "twist – to reset" type. In addition a "Power ON" indicator lamp shall be fitted.
B.	Master Panel shall be fitted internally with an audible and visual alarm which operates at all times the "Open" button is pressed and for a safe period before the leaf is in motion under power. An interlock switch is fitted to the internal access panels for the drive units to prevent any leaf being used it is open. On operation OPEN/CLOSE push button, the variable speed drive control panel shall enable the doors to be started and accelerated to their running speed by a smooth & gradual increase of the speed over an interval of time. Similarly, on receiving a confirmation from the „Pre-final" limit/proximity switch, the panel shall automatically decrease the speed of the door smoothly & gradually from running speed to full rest.

C.	MANUAL OPERATION	A mechanism to ensure two persons able to operate the drivemanually with "D" handle and doors shall be opened by disengaging the Gear Box b y dislocating the chain.
D.	MOTORIZED DRIVE PARAMETERS	The drive system is required to be designed for electrical as well as manual operation, in case of power failure. Change over from electrical to manual operation is to be achieved by shifting the idler gear into manual position and using the "D" type handle provided.
E.	ELECTRICAL CONTROL SYSTEM	The electrical control system shall be able to control allaspects of the door movements. The control system for eachdoor shall consist of panel, limit/proximity switches, trailing/bunching cable etc.
F.	ELECTRICAL POWER SUPPLY	The incoming power supply to the door panel shall be 415V, 3phase AC & 50Cys, 4 wire & earth. Power supply to each master leaf shall be by means of a looped trailing/bunching copper cable 4 Core 6 Sqmm unarmored system, mounted in the top guide track area from a junction box placed at the topextremities inside the hangar.
G.	CONTROL PANEL SYSTEM	Each Door Control Panel shall consist of DOL starters, drive contactors, motor protection, control fuse holder, fused and indication lamps interlock contactors, limit/proximity switches, warning lights/hooters, trailing/bunching cable/wiring on the door after the main incoming power junction box. The incoming and of each control panel will be of separate MCCB 415 Volts,
H.	DOOR OPERATION SAFETY	The following features shall be incorporated in the controlsystem of the door:
	(a)	The Control Panel shall be located as directed for easy operation and enable the operator to have a full view of thearea during operation of the door.
	(b)	The OPEN & CLOSE Push Buttons are constant pressure (HOLD to RUN) and the EMERGENCY STOP Push Buttons isSTAY PUT type and have to be twisted to release.
	(c)	Emergency limit switch shall be provided at the extreme limitsof travel just before the mechanical buffers. These emergency limit switches shall totally cut-off the control/power supply to the system in case the doors over shoot their normal limits.
	(d)	Audio/Visual Alarm: Master door panel shall be provided witha Flash Light cum Hooter that shall start & continue to operate during the time door is in operation.
	(e)	Key operation: An operation key switch shall be provided oneach control panel to prevent unauthorized operation of the doors.